

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q19-1251

Miles Avenue 230 kV
April 23, 2021

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q19-1251. The Interconnection Customer proposes a 500.0 MW battery energy storage facility interconnecting at the Miles Avenue 230 kV substation in the Illinois area. The request was received on July 17, 2019. The study evaluated system intact and single contingency conditions using the regional planning case, including 360 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 1 overloaded facility. The total estimated cost of the network upgrades allocated to this project is \$24,725,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Miles Avenue / Pueblo Xtra 230kV	Co Line Rd / Brunner Island 500kV	124.9	534	100.0

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Miles Avenue / Pueblo Xtra 230kV	Rebuild line with higher-capacity conductor	\$24,725,000	100% (\$24,725,000)
	Total allocated to project		\$24,725,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.